SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

Creation Date 28-Jul-2020 **Revision Date** 09-Feb-2024 **Revision Number** 4

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	**Dimethylamine, 1-2M (5 -10 w/w%) solution in 1-Methyl-2-pyrrolidinone**
Cat No. :	**469460000**
Molecular Formula	C2 H7 N

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company .

EU entity/business name
Thermo Fisher Scientific
Janssen Pharmaceuticalaan 3a, 2440 Geel,
Belgium

UK entity/business name
Fisher Scientific UK
Bishop Meadow Road,
Loughborough, Leicestershire LE11 5RG,
United Kingdom

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

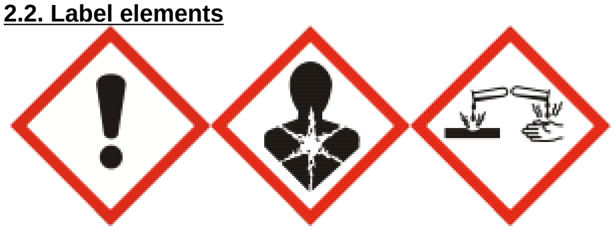
Health hazards

Skin Corrosion/Irritation	Category 1 B (H314)
Serious Eye Damage/Eye Irritation	Category 1 (H318)
Reproductive Toxicity	Category 1B (H360D)
Specific target organ toxicity - (single exposure)	Category 3 (H335)

Environmental hazards
Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word **Danger**

Hazard Statements
H314 - Causes severe skin burns and eye damage
H335 - May cause respiratory irritation
H360D - May damage the unborn child

Precautionary Statements
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician

Additional EU labelling
Restricted to professional users

2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008

Dimethylamine	124-40-3	EEC No. 204-697-4	5-10	Flam. Liq. 1 (H224) Acute Tox. 4 (H302) Acute Tox. 4 (H332) Skin Corr. 1B (H314) Eye Dam. 1 (H318) STOT SE 3 (H335) Aquatic Chronic 3 (H412)
1-Methyl-2-pyrrolidone	872-50-4	EEC No. 212-828-1	90-95	Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) Repr. 1B (H360D) STOT SE 3 (H335)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Dimethylamine	STOT SE 3 :: C>=5%	-	-
1-Methyl-2-pyrrolidone	STOT SE 3 (H335) :: C>=10%	-	-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

Ingestion Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.

Inhalation If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.

Self-Protection of the First Aider Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
CO_2, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

Hazardous Combustion Products
None under normal use conditions.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

6.2. Environmental precautions

Should not be released into the environment.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany)	Storage Class/LGK 6.1D
Switzerland - Storage of hazardous substances	Storage class - SC 6.1 https://www.kvu.ch/de/themen/stoffe-und-produkte https://www.kvu.ch/fr/themes/substances-et-produits https://www.kvu.ch/it/temi/sostanze-e-prodotti

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits
List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
Dimethylamine	TWA: 2 ppm (8h) TWA: 3.8 mg/m³ (8h) STEL: 5 ppm (15min) STEL: 9.4 mg/m³ (15min)	STEL: 6 ppm 15 min STEL: 11 mg/m³ 15 min TWA: 2 ppm 8 hr TWA: 3.8 mg/m³ 8 hr	TWA / VME: 1 ppm (8 heures). restrictive limit TWA / VME: 1.9 mg/m³ (8 heures). restrictive limit STEL / VLCT: 2 ppm. restrictive limit STEL / VLCT: 3.8 mg/m³. restrictive limit	TWA: 2 ppm 8 uren TWA: 3.8 mg/m³ 8 uren STEL: 5 ppm 15 minuten STEL: 9.4 mg/m³ 15 minuten	STEL / VLA-EC: 5 ppm (15 minutos). STEL / VLA-EC: 9.4 mg/m³ (15 minutos). TWA / VLA-ED: 2 ppm (8 horas) TWA / VLA-ED: 3.8 mg/m³ (8 horas)
1-Methyl-2-pyrrolidone	TWA: 40 mg/m³ (8h) TWA: 10 ppm (8h) Skin STEL: 20 ppm (15min) STEL: 80 mg/m³ (15min) STEL: 80 mg/m³ (8h) STEL: 20 ppm (8h)	STEL: 20 ppm 15 min STEL: 80 mg/m³ 15 min TWA: 10 ppm 8 hr TWA: 40 mg/m³ 8 hr Skin	TWA / VME: 40 mg/m³ (8 heures). indicative limit TWA / VME: 10 ppm (8 heures). indicative limit STEL / VLCT: 80 mg/m³. indicative limit STEL / VLCT: 20 ppm. indicative limit Peau	TWA: 10 ppm 8 uren TWA: 40 mg/m³ 8 uren STEL: 20 ppm 15 minuten STEL: 80 mg/m³ 15 minuten Huid	STEL / VLA-EC: 20 ppm (15 minutos). STEL / VLA-EC: 80 mg/m³ (15 minutos). TWA / VLA-ED: 10 ppm (8 horas) TWA / VLA-ED: 40 mg/m³ (8 horas) Piel

Component	Italy	Germany	Portugal	The Netherlands	Finland
Dimethylamine	TWA: 2 ppm 8 ore. Time Weighted Average TWA: 3.8 mg/m³ 8 ore. Time Weighted Average STEL: 5 ppm 15 minuti. Short-term STEL: 9.4 mg/m³ 15 minuti. Short-term	TWA: 2 ppm (8 Stunden). AGW - exposure factor 2 TWA: 3.7 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 2 ppm (8 Stunden). MAK even if the MAK value is adhered to, "odor-associated" symptoms cannot be ruled out in individual cases TWA: 3.7 mg/m³ (8 Stunden). MAK even if the MAK value is adhered to, "odor-associated" symptoms cannot be ruled out in individual cases Höhepunkt: 4 ppm Höhepunkt: 7.4 mg/m³	STEL: 5 ppm 15 minutos STEL: 9.4 mg/m³ 15 minutos TWA: 2 ppm 8 horas TWA: 3.8 mg/m³ 8 horas	TWA: 1.8 mg/m³ 8 uren	TWA: 2 ppm 8 tunteina TWA: 3.7 mg/m³ 8 tunteina STEL: 5 ppm 15 minuutteina STEL: 9.4 mg/m³ 15 minuutteina
1-Methyl-2-pyrrolidone	TWA: 10 ppm 8 ore. Time Weighted Average TWA: 40 mg/m³ 8 ore. Time Weighted Average STEL: 20 ppm 15 minuti. Short-term STEL: 80 mg/m³ 15 minuti. Short-term	TWA: 20 ppm (8 Stunden). AGW - exposure factor 2 TWA: 82 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 20 ppm (8 Stunden). MAK can	STEL: 20 ppm 15 minutos STEL: 80 mg/m³ 15 minutos TWA: 10 ppm 8 horas TWA: 40 mg/m³ 8 horas Pele	huid STEL: 80 mg/m³ 15 minuten TWA: 40 mg/m³ 8 uren	TWA: 3.5 ppm 8 tunteina TWA: 14 mg/m³ 8 tunteina STEL: 20 ppm 15 minuutteina STEL: 80 mg/m³ 15 minuutteina

	Pelle	occur as vapor and aerosol at the same time TWA: 82 mg/m³ (8 Stunden). MAK can occur as vapor and aerosol at the same time Höhepunkt: 40 ppm Höhepunkt: 164 mg/m³ Haut			Iho

Component	Austria	Denmark	Switzerland	Poland	Norway
Dimethylamine	MAK-KZGW: 2 ppm 15 Minuten MAK-KZGW: 3.8 mg/m³ 15 Minuten MAK-TMW: 2 ppm 8 Stunden MAK-TMW: 3.8 mg/m³ 8 Stunden Ceiling: 2 ppm Ceiling: 3.8 mg/m³	TWA: 2 ppm 8 timer TWA: 3.8 mg/m³ 8 timer STEL: 9.4 mg/m³ 15 minutter STEL: 5 ppm 15 minutter	STEL: 4 ppm 15 Minuten STEL: 8 mg/m³ 15 Minuten TWA: 2 ppm 8 Stunden TWA: 4 mg/m³ 8 Stunden	STEL: 9 mg/m³ 15 minutach TWA: 3 mg/m³ 8 godzinach	TWA: 2 ppm 8 timer TWA: 4 mg/m³ 8 timer STEL: 4 ppm 15 minutter. value calculated STEL: 8 mg/m³ 15 minutter. value calculated
1-Methyl-2-pyrrolidone	Haut MAK-KZGW: 7.2 ppm 15 Minuten MAK-KZGW: 28.8 mg/m³ 15 Minuten MAK-TMW: 3.6 ppm 8 Stunden MAK-TMW: 14.4 mg/m³ 8 Stunden	TWA: 5 ppm 8 timer TWA: 20 mg/m³ 8 timer STEL: 80 mg/m³ 15 minutter STEL: 20 ppm 15 minutter Hud	Haut/Peau STEL: 40 ppm 15 Minuten STEL: 160 mg/m³ 15 Minuten TWA: 20 ppm 8 Stunden TWA: 80 mg/m³ 8 Stunden	STEL: 80 mg/m³ 15 minutach TWA: 40 mg/m³ 8 godzinach	TWA: 5 ppm 8 timer TWA: 20 mg/m³ 8 timer STEL: 20 ppm 15 minutter. value from the regulation STEL: 80 mg/m³ 15 minutter. value from the regulation Hud

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Dimethylamine	TWA: 2 ppm TWA: 3.8 mg/m³ STEL : 5 ppm STEL : 9.4 mg/m³	kože TWA-GVI: 2 ppm 8 satima. TWA-GVI: 3.8 mg/m³ 8 satima. STEL-KGVI: 5 ppm 15 minutama. STEL-KGVI: 9.4 mg/m³ 15 minutama.	TWA: 2 ppm 8 hr. TWA: 3.8 mg/m³ 8 hr. STEL: 5 ppm 15 min STEL: 9.4 mg/m³ 15 min	STEL: 5.0 ppm STEL: 9.4 mg/m³ TWA: 2 ppm TWA: 3.8 mg/m³	TWA: 3.8 mg/m³ 8 hodinách. Potential for cutaneous absorption Ceiling: 9 mg/m³
1-Methyl-2-pyrrolidone	TWA: 10 ppm TWA: 40 mg/m³ STEL : 20 ppm STEL : 80 mg/m³ Skin notation	kože TWA-GVI: 10 ppm 8 satima. TWA-GVI: 40 mg/m³ 8 satima. STEL-KGVI: 20 ppm 15 minutama. STEL-KGVI: 80 mg/m³ 15 minutama.	TWA: 10 ppm 8 hr. TWA: 40 mg/m³ 8 hr. STEL: 20 ppm 15 min STEL: 80 mg/m³ 15 min Skin	Skin-potential for cutaneous absorption STEL: 80 mg/m³ STEL: 20 ppm TWA: 40 mg/m³ TWA: 10 ppm	TWA: 40 mg/m³ 8 hodinách. Potential for cutaneous absorption Ceiling: 80 mg/m³ toxic for reproduction

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Dimethylamine	TWA: 2 ppm 8 tundides. TWA: 3.8 mg/m³ 8 tundides. STEL: 5 ppm 15 minutites. STEL: 9.4 mg/m³ 15 minutites.	TWA: 2 ppm 8 hr TWA: 3.8 mg/m³ 8 hr STEL: 5 ppm 15 min STEL: 9.4 mg/m³ 15 min	STEL: 15 ppm STEL: 27 mg/m³ TWA: 10 ppm TWA: 18 mg/m³	STEL: 9.4 mg/m³ 15 percekben. CK TWA: 3.8 mg/m³ 8 órában. AK lehetséges borön keresztüli felszívódás	STEL: 5 ppm STEL: 9.4 mg/m³ TWA: 2 ppm 8 klukkustundum. TWA: 3.8 mg/m³ 8 klukkustundum.
1-Methyl-2-pyrrolidone	Nahk TWA: 10 ppm 8 tundides. TWA: 40 mg/m³ 8 tundides. STEL: 20 ppm 15 minutites. STEL: 80 mg/m³ 15 minutites.	Skin notation TWA: 40 mg/m³ 8 hr TWA: 10 ppm 8 hr STEL: 80 mg/m³ 15 min STEL: 20 ppm 15 min	skin - potential for cutaneous absorption STEL: 20 ppm STEL: 80 mg/m³ TWA: 10 ppm TWA: 40 mg/m³	STEL: 80 mg/m³ 15 percekben. CK TWA: 40 mg/m³ 8 órában. AK lehetséges borön keresztüli felszívódás	STEL: 20 ppm STEL: 80 mg/m³ TWA: 10 ppm 8 klukkustundum. TWA: 40 mg/m³ 8 klukkustundum.

Component	Latvia	Lithuania	Luxembourg	Malta	Romania

Dimethylamine	STEL: 5 ppm STEL: 9.4 mg/m³ TWA: 2 ppm TWA: 3.8 mg/m³	TWA: 2 ppm IPRD TWA: 3.8 mg/m³ IPRD STEL: 5 ppm STEL: 9.4 mg/m³	TWA: 2 ppm 8 Stunden TWA: 3.8 mg/m³ 8 Stunden STEL: 5 ppm 15 Minuten STEL: 9.4 mg/m³ 15 Minuten	TWA: 2 ppm TWA: 3.8 mg/m³ STEL: 5 ppm 15 minuti STEL: 9.4 mg/m³ 15 minuti	TWA: 2 ppm 8 ore TWA: 3.8 mg/m³ 8 ore STEL: 9.4 mg/m³ 15 minute STEL: 5 ppm 15 minute
1-Methyl-2-pyrrolidone	skin - potential for cutaneous exposure STEL: 20 ppm STEL: 80 mg/m³ TWA: 10 ppm TWA: 40 mg/m³	TWA: 10 ppm IPRD TWA: 40 mg/m³ IPRD Oda STEL: 20 ppm STEL: 80 mg/m³	Possibility of significant uptake through the skin TWA: 40 mg/m³ 8 Stunden TWA: 10 ppm 8 Stunden STEL: 80 mg/m³ 15 Minuten STEL: 20 ppm 15 Minuten	possibility of significant uptake through the skin TWA: 40 mg/m³ TWA: 10 ppm STEL: 80 mg/m³ 15 minuti STEL: 20 ppm 15 minuti	Skin notation TWA: 10 ppm 8 ore TWA: 40 mg/m³ 8 ore STEL: 20 ppm 15 minute STEL: 80 mg/m³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Dimethylamine	Skin notation MAC: 1 mg/m³	Ceiling: 9.4 mg/m³ TWA: 2 ppm TWA: 3.8 mg/m³	TWA: 2 ppm 8 urah TWA: 3.8 mg/m³ 8 urah STEL: 5 ppm 15 minutah STEL: 9.4 mg/m³ 15 minutah	Binding STEL: 5 ppm 15 minuter Binding STEL: 9 mg/m³ 15 minuter TLV: 2 ppm 8 timmar. NGV TLV: 3.5 mg/m³ 8 timmar. NGV	TWA: 2 ppm 8 saat TWA: 3.8 mg/m³ 8 saat STEL: 5 ppm 15 dakika STEL: 9.4 mg/m³ 15 dakika
1-Methyl-2-pyrrolidone	MAC: 100 mg/m³	Ceiling: 80 mg/m³ Potential for cutaneous absorption TWA: 40 mg/m³ TWA: 10 ppm	TWA: 10 ppm 8 urah vapor TWA: 40 mg/m³ 8 urah vapor Koža STEL: 20 ppm 15 minutah vapor STEL: 80 mg/m³ 15 minutah vapor	Binding STEL: 20 ppm 15 minuter Binding STEL: 80 mg/m³ 15 minuter TLV: 3.6 ppm 8 timmar. NGV TLV: 14.4 mg/m³ 8 timmar. NGV Hud	Deri TWA: 10 ppm 8 saat TWA: 40 mg/m³ 8 saat STEL: 20 ppm 15 dakika STEL: 80 mg/m³ 15 dakika

Biological limit values
List source(s):

Component	European Union	United Kingdom	France	Spain	Germany
1-Methyl-2-pyrrolidone				2-Hydroxy-N-methylsuccinimide: 20 mg/g Creatinine urine pre-shift 5-Hydroxy-N-methyl-2-pyrrolidone: 70 mg/g Creatinine urine between 2-4 hours after the final exposure	5-Hydroxy-N-methyl-2-pyrrolidone: 150 mg/L urine (end of shift)

Component	Italy	Finland	Denmark	Bulgaria	Romania
1-Methyl-2-pyrrolidone		5-Hydroxy-N-methyl-2-pyrrolidone: 8 µmol/mol Creatinine urine in the morning after a working day. 2-Hydroxy-N-methyl-succinimide: 5 µmol/mol Creatinine urine after the shift.			

Monitoring methods
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Dimethylamine 124-40-3 (5-10)		DNEL = 1.95mg/kg bw/day		DNEL = 0.0874mg/kg bw/day
1-Methyl-2-pyrrolidone 872-50-4 (90-95)				DNEL = 4.8mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Dimethylamine 124-40-3 (5-10)	DNEL = 12.9mg/m³	DNEL = 9.4mg/m³		DNEL = 3.8mg/m³
1-Methyl-2-pyrrolidone 872-50-4 (90-95)			DNEL = 40mg/m³	DNEL = 14.4mg/m³

Predicted No Effect Concentration (PNEC)
See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Dimethylamine 124-40-3 (5-10)	PNEC = 0.06mg/L	PNEC = 3.26mg/kg sediment dw	PNEC = 0.06mg/L	PNEC = 100mg/L	PNEC = 0.0385mg/kg soil dw
1-Methyl-2-pyrrolidone 872-50-4 (90-95)	PNEC = 0.25mg/L	PNEC = 1.09mg/kg sediment dw	PNEC = 5mg/L	PNEC = 10mg/L	PNEC = 0.0701mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Dimethylamine 124-40-3 (5-10)	PNEC = 0.006mg/L	PNEC = 0.33mg/kg sediment dw			
1-Methyl-2-pyrrolidone 872-50-4 (90-95)	PNEC = 0.025mg/L	PNEC = 0.109mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures
Ensure that eyewash stations and safety showers are close to the workstation location.
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Neoprene	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

	To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly
Large scale/emergency use	Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced **Recommended Filter type:** Inorganic gases and vapours filter Type B Grey or Ammonia and organic ammonia derivatives filter Type K Green conforming to EN14387
Small scale/Laboratory use	Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. **Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141 When RPE is used a face piece Fit Test should be conducted
Environmental exposure controls	No information available.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Pale yellow	
Odor	No information available	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	No information available	**Method -** No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	No data available	
Water Solubility	No information available	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
Dimethylamine	-0.274	
1-Methyl-2-pyrrolidone	-0.46	
Vapor Pressure	No data available	
Density / Specific Gravity	0.996	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	(liquid) Not applicable	

9.2. Other information

Molecular Formula	C2 H7 N
Molecular Weight	45.08

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat.

10.5. Incompatible materials

None known.

10.6. Hazardous decomposition products

None under normal use conditions.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;
Oral Based on available data, the classification criteria are not met
Dermal Based on available data, the classification criteria are not met
Inhalation Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Dimethylamine	LD50 = 698 mg/kg (Rat)	LD50 = 3900 mg/kg (Rat)	LC50 = 7340 ppm (Rat) 20 min
1-Methyl-2-pyrrolidone	LD50 = 3914 mg/kg (Rat)	LD50 = 8 g/kg (Rabbit)	LC50 > 5.1 mg/L (Rat) 4 h

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;
Respiratory No data available
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Category 1B
Reproductive Effects May cause harm to the unborn child.

(h) STOT-single exposure; Category 3

Results / Target organs Respiratory system.

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

Symptoms / effects,both acute and delayed Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity
Ecotoxicity effects .

Component	Freshwater Fish	Water Flea	Freshwater Algae
Dimethylamine	LC50: = 396 mg/L, 96h static (Brachydanio rerio) LC50: 127 - 349 mg/L, 96h semi-static (Poecilia reticulata) LC50: = 210 mg/L, 96h static (Poecilia reticulata) LC50: = 120 mg/L, 96h static (Oncorhynchus mykiss) LC50: 111 - 125 mg/L, 96h (Oncorhynchus mykiss)	EC50: = 88.7 mg/L, 48h (Daphnia magna Straus)	EC50: = 9 mg/L, 96h (Pseudokirchneriella subcapitata)
1-Methyl-2-pyrrolidone	LC50: = 1400 mg/L, 96h static (Poecilia reticulata) LC50: = 1072 mg/L, 96h static (Pimephales promelas) LC50: = 832 mg/L, 96h static (Lepomis macrochirus)	EC50: = 4897 mg/L, 48h (Daphnia magna)	EC50: > 500 mg/L, 72h (Desmodesmus subspicatus)

12.2. Persistence and degradability
No information available

Persistence Persistence is unlikely.

Component	Degradability
1-Methyl-2-pyrrolidone 872-50-4 (90-95)	water: 73% 28 days OECD 301C soil: >=90% 21 days

12.3. Bioaccumulative potential
Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Dimethylamine	-0.274	No data available
1-Methyl-2-pyrrolidone	-0.46	No data available

12.4. Mobility in soil
No information available

12.5. Results of PBT and vPvB assessment
No data available for assessment.

12.6. Endocrine disrupting properties
Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant	This product does not contain any known or suspected substance
Ozone Depletion Potential	This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products	Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.
Contaminated Packaging	Dispose of this container to hazardous or special waste collection point.
European Waste Catalogue (EWC)	According to the European Waste Catalog, Waste Codes are not product specific, but application specific.
Other Information	Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.
Switzerland - Waste Ordinance	Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600 https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number	UN3267
14.2. UN proper shipping name	CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.
Technical Shipping Name	Dimethylamine in 1-Methyl-2-pyrrolidinone
14.3. Transport hazard class(es)	8
14.4. Packing group	II

ADR

14.1. UN number	UN3267
14.2. UN proper shipping name	CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.
Technical Shipping Name	Dimethylamine in 1-Methyl-2-pyrrolidinone
14.3. Transport hazard class(es)	8
14.4. Packing group	II

IATA

14.1. UN number	UN3267
14.2. UN proper shipping name	CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.
Technical Shipping Name	Dimethylamine in 1-Methyl-2-pyrrolidinone
14.3. Transport hazard class(es)	8
14.4. Packing group	II

14.5. Environmental hazards	No hazards identified
14.6. Special precautions for user	No special precautions required.
14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Dimethylamine	124-40-3	204-697-4	-	-	X	X	KE-11124	X	X
1-Methyl-2-pyrrolidone	872-50-4	212-828-1	-	-	X	X	KE-25324	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Dimethylamine	124-40-3	X	ACTIVE	X	-	X	X	X
1-Methyl-2-pyrrolidone	872-50-4	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Dimethylamine	124-40-3	-	Use restricted. See item 75. (see link for restriction details)	-
1-Methyl-2-pyrrolidone	872-50-4	-	Use restricted. See item 72. (see link for restriction details) Use restricted. See item 30. (see link for restriction details) Use restricted. See item 71. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	SVHC Candidate list - 212-828-1 - Toxic for reproduction, Article 57c

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

REACH links
https://echa.europa.eu/authorisation-list
https://echa.europa.eu/substances-restricted-under-reach
https://echa.europa.eu/candidate-list-table

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Dimethylamine	124-40-3	Not applicable	Not applicable
1-Methyl-2-pyrrolidone	872-50-4	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and

import of dangerous chemicals
Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?
Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values
Take note of Directive 94/33/EC on the protection of young people at work
Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 1 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Dimethylamine	WGK1	Class I : 20 mg/m³ (Massenkonzentration)
1-Methyl-2-pyrrolidone	WGK1	

Component	France - INRS (Tables of occupational diseases)
Dimethylamine	Tableaux des maladies professionnelles (TMP) - RG 49,RG 49bis
1-Methyl-2-pyrrolidone	Tableaux des maladies professionnelles (TMP) - RG 84

Swiss Regulations
Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
1-Methyl-2-pyrrolidone 872-50-4 (90-95)		Group I	

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H335 - May cause respiratory irritation
H360D - May damage the unborn child
H224 - Extremely flammable liquid and vapor
H302 - Harmful if swallowed
H315 - Causes skin irritation
H319 - Causes serious eye irritation
H332 - Harmful if inhaled
H412 - Harmful to aquatic life with long lasting effects

Legend

CAS - Chemical Abstracts Service
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
PICCS - Philippines Inventory of Chemicals and Chemical Substances
ENCS - Japanese Existing and New Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
AICS - Australian Inventory of Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
TWA - Time Weighted Average
ACGIH - American Conference of Governmental Industrial Hygienists
IARC - International Agency for Research on Cancer
DNEL - Derived No Effect Level
Predicted No Effect Concentration (PNEC)
RPE - Respiratory Protective Equipment
LD50 - Lethal Dose 50%
LC50 - Lethal Concentration 50%
EC50 - Effective Concentration 50%
NOEC - No Observed Effect Concentration
POW - Partition coefficient Octanol:Water
PBT - Persistent, Bioaccumulative, Toxic
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
MARPOL - International Convention for the Prevention of Pollution from Ships
OECD - Organisation for Economic Co-operation and Development
ATE - Acute Toxicity Estimate
BCF - Bioconcentration factor
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.
Chemical incident response training.

Creation Date	28-Jul-2020
Revision Date	09-Feb-2024
Revision Summary	Not applicable.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer
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End of Safety Data Sheet